Can disease spread be modelled by a diffusion equation?

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Abstract

In this study, we modelled the spread of disease in Manchester through a reaction-diffusion equation, comparing this to pre-lockdown Covid data. The main assumptions of the model were a constant population density, a recovery time of 14 days, and effects on immunocompromised people is ignored. The Euler method is used to solve the equation computationally. Our study successfully/unsuccessfully modelled spatial disease spread when comparing to pre-pandemic Covid data, but only the initial spread. The model could be improved by implementing random-walks to model more random behaviour, modelling more outbreak centres, or accounting for disease spread reduction.

Introduction

Modelling disease spread is difficult, want better predicitions so governments can make plans.

When first thinking about this problem, we considered x y z.

Theory - Reaction-Diffusion Equation

We attempt to model the spread of disease via a reaction-diffusion equation,

$$\frac{\partial I}{\partial t} = D\nabla^2 I + \beta(x,y)SI - \gamma I, \tag{1}$$

which models spatial densities ρ, which are parametrised such that,

$$\rho = S + I + R, \tag{2}$$

where,

- $S(x,y,t)$: Susceptible population.
  - Those who can catch disease.
- $I(x,y,t)$: Infected population.
  - Those who have the disease.
- $R(x,y,t)$: Recovered population.
  - Those who are no longer infected.

Results: Simulation vs Data

Put things here

Results: Figures

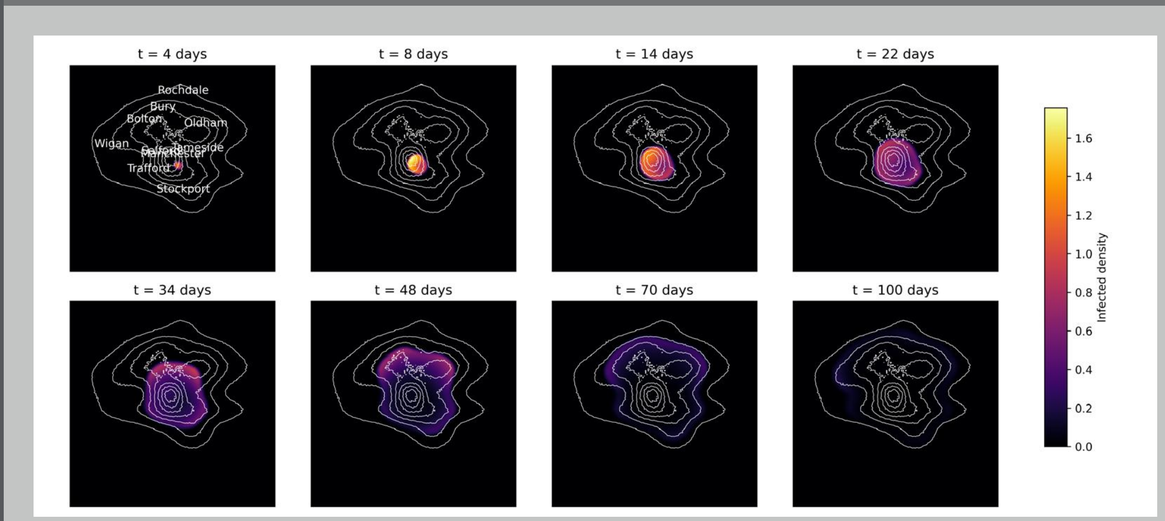

Figure 1: Disease spreading over time in Greater Manchester.

Discussion

This worked/didn't work. This will for sure effect something.

Conclusions

We things out ig.

References